

Relation between h' and zero free region of $L(s, \chi)$

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Abstract. Siegel's theorem gives us that the class number $h(-d)$ of an imaginary quadratic field of discriminant $-d$ satisfies the ineffective lower bound $h(-d) \gg_{\epsilon} d^{\frac{1}{2}-\epsilon}$ for every $\epsilon > 0$. We will follow the arguments of Mahler[1] to see how this lower bound can be improved given a zero free region of $L(s, \chi_d)$, using a formula for ζ_Q , due to Mordel[2].

Notation

Reduced Binary quadratic forms of discriminant $-d$ will be denoted by Q . In section 1 the discriminant is denoted by $d < 0$ and in section 2 the discriminant is denoted by $-d < 0$. $\sum_Q \cdot$ is used to denote the summation over reduced Binary quadratic forms with given discriminant.

1 Relation between ζ_Q and quadratic Dedekind Zeta function

For a reduced binary quadratic form with discriminant $d < 0$, define the representation number of n by Q as

$$r_Q(n) := \frac{1}{2} \# \{(x, y) = 1 : Q(x, y) = n\}.$$

Define

$$r_d(n) := \sum_Q r_Q(n),$$

We will be using the following theorem without proof.

Theorem 1. *Let $d < 0$ be a fundamental discriminant. Then*

$$r_d(n) = \begin{cases} 0 & \text{if } p^2 \mid n \text{ and } p \mid d, \\ \prod_{\substack{p \mid n \\ p \nmid d}} (1 + \chi_d(p)) & \text{otherwise.} \end{cases}$$

Lemma 1.

$$\sum_{n=1}^{\infty} \frac{r_d(n)}{n^s} = \prod_p \left(\frac{1 + p^{-s}}{1 - \chi_d(p)p^{-s}} \right), \quad \text{where } \chi_d(n) = \left(\frac{d}{n} \right).$$

Proof. Note that

$$\frac{1 + p^{-s}}{1 - \chi_d(p)p^{-s}} = \begin{cases} 1 & \text{if } \chi_d(p) = -1, \\ 1 + p^{-s} & \text{if } \chi_d(p) = 0, \\ 1 + 2 \sum_{k=1}^{\infty} p^{-ks} & \text{if } \chi_d(p) = 1. \end{cases}$$

Thus,

$$\prod_p \frac{1 + p^{-s}}{1 - \chi_d(p)p^{-s}} = \prod_{\chi_d(p)=-1} 1 \prod_{\chi_d(p)=0} \left(1 + \frac{1}{p^s}\right) \prod_{\chi_d(p)=1} \left(1 + \frac{2}{p^s} + \frac{2}{p^{2s}} + \frac{2}{p^{3s}} \dots\right).$$

Case I: $p^2 \mid n$ and $p \mid d$. Then $\chi_d(p) = 0$, so the p^{-2s} term never appears in the product. Hence such n do not appear in the Dirichlet series.

Case II: Otherwise.

The coefficient of n^{-s} in $r_d(n)$ is

$$r_d(n) = \prod_{\substack{p \mid n \\ p \nmid d}} (1 + \chi_d(p)).$$

Since, if $\chi_d(p) = -1$ for $p \mid n$, then the p^{-s} term never occurs in the product, so $r_d(n) = 0$. If $\chi_d(p) \neq -1$ for all $p \mid n$, then:

- for $\chi_d(p) = 1$, we get a contribution $2/p^{ks}$ ($p^k \mid n$),
- for $\chi_d(p) = 0$, we get $1/p^s$ (Note that $p \mid n$, otherwise it would be case I),

Hence the coefficient matches $r_d(n)$, proving the lemma. □

Theorem 2.

$$\sum_Q \zeta_Q(s) = \zeta(s) L(s, \chi_d).$$

Proof.

$$\begin{aligned} \sum_Q \zeta_Q(s) &= \sum_Q \frac{1}{2} \sum_{(m,n) \neq (0,0)} Q(m,n)^{-s} \\ &= \sum_Q \frac{1}{2} \sum_{m=1}^{\infty} m^{-2s} \sum_{(u,v)=1} Q(u,v)^{-s} \\ &= \sum_Q \left(\sum_{m=1}^{\infty} m^{-2s} \right) \left(\frac{1}{2} \sum_{(u,v)=1} Q(u,v)^{-s} \right) \\ &= \sum_Q \zeta(2s) \sum_{n=1}^{\infty} \frac{r_Q(n)}{n^s} \\ &= \zeta(2s) \sum_{n=1}^{\infty} \frac{r_d(n)}{n^s}. \end{aligned}$$

Now consider the Euler product:

$$\begin{aligned}
\zeta(2s)^{-1}\zeta(s)L(s, \chi_d) &= \prod_p (1 - p^{-2s}) \prod_p \frac{1}{1 - p^{-s}} \prod_p \frac{1}{1 - \chi_d(p)p^{-s}} \\
&= \prod_p \frac{1 - p^{-2s}}{(1 - p^{-s})(1 - \chi_d(p)p^{-s})} \\
&= \prod_p \frac{1 + p^{-s}}{1 - \chi_d(p)p^{-s}}.
\end{aligned}$$

By Lemma 1,

$$\prod_p \frac{1 + p^{-s}}{1 - \chi_d(p)p^{-s}} = \sum_{n=1}^{\infty} \frac{r_d(n)}{n^s}.$$

Thus,

$$\zeta(s)L(s, \chi_d) = \zeta(2s) \sum_{n=1}^{\infty} \frac{r_d(n)}{n^s} = \sum_Q \zeta_Q(s).$$

□

2 Formula for ζ_Q

Let Q be a reduced binary quadratic form

$$Q(x, y) = ax^2 + bxy + cy^2,$$

with $a > 0$ and discriminant $b^2 - 4ac = -d < 0$.

Lemma 2.

$$I(g) := \int_{-\infty}^{\infty} \frac{e^{2\pi i g z}}{(z^2 + 1)^s} dz,$$

where $g = \frac{mn\sqrt{d}}{2a}$, $m \in \mathbb{Z}_{>0}$, $n \in \mathbb{Z}_{>0}$. Then for $\sigma \geq \frac{1}{2}$,

$$I(g) \ll e^{-\pi g}.$$

Proof. Note that $g = \frac{mn\sqrt{d}}{2a} \geq \frac{\sqrt{3}}{2}$, as $a \leq \sqrt{|d|/3}$.

We split $I(g)$ into three integrals:

$$I(g) = I_1 + I_2 + I_3,$$

where I_1 is over the upper half-circle centered at origin, I_2 over the circle $|z - i| = \frac{1}{2}$, and I_3 over the two sides of the line $x + i$, $x \in [\frac{1}{2}, \infty)$ running in opposite direction.

For I_1 , we have

$$I_1 = \lim_{R \rightarrow \infty} \int_0^\pi \frac{e^{2\pi i g R e^{i\theta}}}{(R^2 e^{2i\theta} + 1)^s} i R e^{i\theta} d\theta = 0,$$

since $\sigma > 0$ and

$$\int_0^\pi e^{-2\pi g R \sin \theta} R d\theta \ll \int_0^{\pi/2} e^{-4g R \theta} R d\theta < \frac{1}{2g}.$$

For I_2 , writing $z = i + \frac{1}{2}e^{i\theta}$, where $0 \leq \theta \leq 2\pi$

$$\left| \frac{e^{2\pi i g z}}{(z^2 + 1)^s} \right| = \frac{e^{-2\pi g - \pi g \sin \theta}}{2^{-s} e^{i\theta s} |z + i|^s} \ll e^{-\pi g}.$$

For I_3 , let $z = i + \ell$, $\ell \geq \frac{1}{2}$. Then for $\sigma > 1/2$

$$\begin{aligned} I_3 &= e^{-2\pi g} \int_{1/2}^\infty \frac{e^{2\pi i g \ell}}{\ell^s (2i + \ell)^s} d\ell, \\ |I_3| &\leq \int_{1/2}^\infty \frac{d\ell}{|\ell^2 + 2i\ell|^\sigma} \ll \int_{1/2}^\infty \frac{d\ell}{\ell^{2\sigma}} \ll 1. \end{aligned}$$

Hence $I_3 \ll e^{-2\pi g}$.

Combining all three estimates,

$$I(g) \ll e^{-\pi g}.$$

□

Lemma 3.

$$R_Q(s) := 2 \sum_{n=1}^\infty \sum_{\substack{m=-\infty \\ m \neq 0}}^\infty \int_{-\infty}^\infty \frac{e^{2\pi i m n \xi}}{(a\xi^2 + b\xi + cn^2)^s} d\xi.$$

Then for $\sigma > 1/2$, we have

$$R_Q(s) \ll \frac{a^{\sigma-1}}{d^{\sigma-\frac{1}{2}}} e^{-\pi\sqrt{d}/(4a)}.$$

Proof. Let

$$I_{m,n} = \int_{-\infty}^\infty \frac{e^{2\pi i m n \xi}}{(a\xi^2 + b\xi + cn^2)^s} d\xi.$$

Then

$$\begin{aligned} I_{m,n} &= \frac{1}{n^{2s-1}} \int_{-\infty}^\infty \frac{e^{2\pi i m n \xi}}{(a\xi^2 + b\xi + c)^s} d\xi \\ &= \frac{a^{s-1}}{n^{2s-1}} \int_{-\infty}^\infty \frac{e^{2\pi i m n \xi}}{\left((a\xi + \frac{b}{2})^2 + \frac{d}{4}\right)^s} d\xi. \end{aligned}$$

After completing the square and change of variables,

$$I_{m,n} = \frac{a^{s-1}}{n^{2s-1}} \left(\frac{4}{d}\right)^{s-1/2} e^{-\pi i m n b/a} I(g),$$

where $I(g)$ is as in Lemma 2. Using Lemma 2,

$$|I_{m,n}| \ll \frac{a^{\sigma-1}}{n^{2\sigma-1}} \frac{1}{d^{\sigma-1/2}} e^{-\pi g}.$$

Summing over m, n ,

$$\begin{aligned} |R_Q(s)| &\ll \sum_{m,n} I_{m,n} \\ &\ll \sum_{m,n} \frac{a^{\sigma-1} e^{-\pi mnd^{1/2}/(2a)}}{d^{\sigma-\frac{1}{2}} n^{2\sigma-1}} \\ &\ll \frac{a^{\sigma-1}}{d^{\sigma-\frac{1}{2}}} e^{-\pi d^{1/2}/(4a)} \sum_{n=1}^{\infty} \frac{1}{n^{2\sigma-1}} \sum_{m=1}^{\infty} e^{-\pi mnd^{1/2}/(4a)} \\ &\ll \frac{a^{\sigma-1}}{d^{\sigma-\frac{1}{2}}} e^{-\pi d^{1/2}/(4a)} \sum_{n=1}^{\infty} \frac{1}{n^{2\sigma-1} (e^{\pi nd^{1/2}/(4a)} - 1)} \\ &\ll \frac{a^{\sigma-1}}{d^{\sigma-\frac{1}{2}}} e^{-\pi d^{1/2}/(4a)} \sum_{n=1}^{\infty} \frac{1}{e^{\pi nd^{1/2}/(4a)} - 1}, \quad \text{as } \sigma \geq \frac{1}{2} \\ &\ll \frac{a^{\sigma-1}}{d^{\sigma-\frac{1}{2}}} e^{-\pi d^{1/2}/(4a)}, \quad \text{as } \sum_{n=1}^{\infty} \frac{1}{e^{\pi nd^{1/2}/(4a)} - 1} \ll 1, \text{ for } \frac{d^{1/2}}{a} \geq \sqrt{3}. \end{aligned}$$

□

In the bound we got for $R_Q(s)$ we have bound for a , $1 \leq a \leq d^{1/2}$. Hence Lemma 3 also gives us a bound just in terms of d , independent of a .

Theorem 3.

$$2\zeta_Q(s) = 2a^{-s}\zeta(2s) + 2d^{\frac{1}{2}-s}a^{s-1} \left(\zeta(2s-1) \frac{\Gamma(s-\frac{1}{2})\sqrt{\pi}}{\Gamma(s)} + O(1) \right), \quad \frac{1}{2} \leq s \leq 1.$$

Proof.

$$\begin{aligned} 2\zeta_Q(s) &= \sum_{x^2+y^2>0} Q(x,y)^{-s} \\ &= \sum_{n=-\infty}^{\infty} \sum_{\substack{m=-\infty \\ (m,n) \neq (0,0)}}^{\infty} \frac{1}{(am^2 + bmn + cn^2)^s}. \end{aligned}$$

Separating $n = 0$,

$$\sum_{m \neq 0} \frac{1}{(am^2)^s} = 2a^{-s}\zeta(2s).$$

For the remaining terms,

$$2 \sum_{n=1}^{\infty} \sum_{m=-\infty}^{\infty} \frac{1}{(am^2 + bmn + cn^2)^s}.$$

Using Poisson summation,

$$\sum_{m=-\infty}^{\infty} f(m) = \sum_{m=-\infty}^{\infty} \int_{-\infty}^{\infty} e^{2\pi imx} f(x) dx.$$

Separating the $m = 0$ term,

$$2 \sum_{n=1}^{\infty} \int_{-\infty}^{\infty} \frac{dx}{(ax^2 + bx + cn^2)^s} = 2 \sum_{n=1}^{\infty} \frac{1}{n^{2s-1}} \int_{-\infty}^{\infty} \frac{dy}{(ay^2 + by + c)^s} \quad (1)$$

$$= 2\zeta(2s-1) \int_{-\infty}^{\infty} \frac{dy}{(ay^2 + by + c)^s} \quad (2)$$

$$= 2d^{1/2-s} a^{s-1} \zeta(2s-1) \beta(s-1/2, 1/2) \quad (3)$$

$$= 2d^{1/2-s} a^{s-1} \zeta(2s-1) \frac{\Gamma(s-\frac{1}{2})\sqrt{\pi}}{\Gamma(s)}. \quad (4)$$

Where $\beta(z_1, z_2)$ denotes the beta function, given by $B(z_1, z_2) = \int_0^1 t^{z_1-1} (1-t)^{z_2-1} dt$, for $z_1, z_2 \in \mathbb{C}$. We get (4) by completing square and change of variables. The remaining terms is exactly $R_Q(s)$, which is bounded by Lemma 3. Hence,

$$2\zeta_Q(s) = 2a^{-s}\zeta(2s) + 2d^{1/2-s}a^{s-1} \left(\zeta(2s-1) \frac{\Gamma(s-\frac{1}{2})\sqrt{\pi}}{\Gamma(s)} + O(1) \right).$$

□

3 Main Theorem

We finally derive a lower bound on h' given a

Lemma 4. $\zeta_K(s) = \zeta(2s)f_d(s) + d^{1/2-s}f_d(1-s) \left(\frac{\zeta(2s-1)\Gamma(s-\frac{1}{2})\sqrt{\pi}}{\Gamma(s)} + a_d(s) \right)$ where $f_d(s) = \sum_Q a^{-s}$

Proof. From theorem 3 we have,

$$\begin{aligned} 2\zeta_Q(s) &= \sum_{x^2+y^2>0} Q(x, y)^{-s} \\ &= 2a^{-s}\zeta(2s) + 2d^{1/2-s}a^{s-1} \left(\frac{\zeta(2s-1)\Gamma(s-\frac{1}{2})\sqrt{\pi}}{\Gamma(s)} + O(1) \right) \end{aligned} \quad (5)$$

Using Theorem 2 we get,

$$\begin{aligned} \zeta_K(s) &= \sum_Q \zeta_Q(s) \\ &= \zeta(2s) \sum_Q a^{-s} + d^{1/2-s} \sum_Q a^{-(1-s)} \left(\frac{\zeta(2s-1)\Gamma(s-\frac{1}{2})\sqrt{\pi}}{\Gamma(s)} + a_d(s) \right) \end{aligned} \quad (6)$$

□

Define

$$A_d(s) = \zeta(2s), \quad B_d(s) = \left(\frac{\zeta(2s-1)\Gamma(s-\frac{1}{2})\sqrt{\pi}}{\Gamma(s)} + a_d(s) \right) (s-1)$$

Then we have the following Lemma

Lemma 5.

$$c_1 \leq A_d(s) \leq c_2, \quad c_6 \leq B_d(s) \leq c_7$$

where c_1, c_2, c_6, c_7 are +ve no's.

Proof. Take an arbitrary constant

$$\frac{1}{2} < \sigma_0 < 1$$

Then $\exists$ positive numbers $c_1, \dots, c_5$ which depend only on σ_0 s.t for sufficiently large d ,

$$c_1 \leq \zeta(2s) \leq c_2, \quad |a_d(s)| \leq c_5$$

note:

$$\frac{\Gamma(s-\frac{1}{2})\sqrt{\pi}}{\Gamma(s)} \geq 0, \quad \zeta(2s-1) \leq 0, \quad s-1 \leq 0$$

so we get

$$\zeta(2s-1) \frac{\Gamma(s-\frac{1}{2})\sqrt{\pi}}{\Gamma(s)} (s-1) \geq 0$$

hence we get positive constant c_3, c_4 such that

$$c_3 \leq \frac{\zeta(2s-1)\Gamma(s-\frac{1}{2})\sqrt{\pi}}{\Gamma(s)} \leq c_4$$

for s in interval $\sigma_0 \leq s \leq 1$, take $\sigma_1 = \max\left(\sigma_0, 1 - \frac{c_3}{2c_5}\right)$ then for $\sigma_1 \leq s \leq 1$

$$-\frac{c_3}{2c_5} \leq s-1 \leq 0$$

$$-\frac{c_3}{2} \leq a_d(s)(s-1) \leq \frac{c_3}{2}$$

hence

$$c_6 = \frac{c_3}{2} \leq B_d(s) \leq c_4 + \frac{c_3}{2} = c_7$$

□

Lemma 6.

$$h'd^{-(1-s)} \leq \frac{f_d(1-s)}{f_d(s)} \leq h'$$

where $h' = \frac{h}{\sum_Q \frac{1}{a}}$, $h = \sum_Q 1$

Proof. Note

$$0 < a \leq \sqrt{\frac{1}{3} \cdot d} \leq \sqrt{d}$$

$$f_d(1-s) = \sum_Q a^{-(1-s)} \leq \sum_Q 1 = h$$

$$\geq \sum_Q d^{-\frac{1}{2}(1-s)}$$

$$f_d(s) = \sum_Q a^{-s} \geq \sum_Q \frac{1}{a}$$

$$\leq \sum_Q a^{-1} d^{\frac{1}{2}(1-s)}$$

thus hence we get

$$h' d^{-(1-s)} \leq \frac{f_d(1-s)}{f_d(s)} \leq h'$$

□

Lemma 7.

$$X_d(s) := \frac{\zeta(s)L(s, \chi_d)(1-s)}{B_d(s)f_d(s)} \leq \frac{c_2}{c_6}(1-s) - \frac{h'}{\sqrt{d}} \quad (*)$$

$$Y_d(s) := \frac{X_d(s)}{d^{1-s}} \geq \frac{c_1}{c_7}(1-s)d^{-(1-s)} - \frac{h'}{\sqrt{d}} \quad (**)$$

for $\sigma_1 \leq s \leq 1$.

Proof. Using the inequalities from lemma 5,6 we get

$$\begin{aligned} X_d(s) &= \frac{\zeta(s)L(s, \chi_d)(1-s)}{B_d(s)f_d(s)} \\ &= \frac{1-s}{B_d(s)f_d(s)} \left(A_d(s)f_d(s) - \frac{B_d(s)}{1-s} f_d(1-s)d^{\frac{1}{2}-s} \right) \\ &= \frac{A_d(s)}{B_d(s)}(1-s) - \frac{f_d(1-s)}{f_d(s)} d^{\frac{1}{2}-s} \\ &\leq \frac{c_2}{c_6}(1-s) - \frac{h'}{\sqrt{d}} \end{aligned}$$

$$\begin{aligned} Y_d(s) &= \frac{X_d(s)}{d^{1-s}} \\ &= \frac{A_d(s)}{B_d(s)}(1-s)d^{-(1-s)} - \frac{f_d(1-s)}{f_d(s)} d^{-\frac{1}{2}} \\ &\geq \frac{c_1}{c_7}(1-s)d^{-(1-s)} - \frac{h'}{\sqrt{d}} \end{aligned}$$

□

Theorem 4. Suppose that γ is a +ve constant and that the integer $d > 0$ is greater than a certain no. d_0 which depends only on γ . Then if the L -series $L(s, \chi)$ has atleast one zero in the interval

$$1 - \frac{\gamma}{\log d} \leq s \leq 1$$

then there exists a number $\Gamma > 0$ depending only on γ such that

$$h' \leq \Gamma \frac{\sqrt{d}}{\log d}$$

Proof. Note

$$\frac{1-s}{B_d(s)f_d(s)} \geq 0$$

hence

$$X_d(s), \zeta(s)L(s, \chi) \text{ have same sign in } J$$

$$\Rightarrow X_d(s), L(s, \chi) \text{ have opp sign in } J \text{ as } \zeta(s) < 0$$

when s is sufficiently close to 1 using (*) we get $-X_d(s) > 0$

$$\Rightarrow L(s, \chi) > 0$$

$$\Rightarrow \frac{c_2}{c_6}(1-s) - \frac{h'}{\sqrt{d}} < 0$$

i.e

$$s > 1 - \frac{c_6}{c_2} \frac{h'}{\sqrt{d}}$$

so if $L(s, \chi)$ has a zero in

$$1 - \frac{\gamma}{\log d} \leq s \leq 1$$

then

$$1 - \frac{\gamma}{\log d} \leq 1 - \frac{c_6}{c_2} \frac{h'}{\sqrt{d}}$$

$$\Rightarrow h' \leq \Gamma \frac{\sqrt{d}}{\log d}$$

□

Theorem 5. Suppose that g is a +ve constant and that the integer $d > 0$ is greater than a certain d_1 which depends only on g . then if the L -series $L(s, \chi)$ has no zero in the interval

$$1 - \frac{g}{\log d} \leq s \leq 1$$

then there exists a number $G > 0$ depending only on g such that

$$h' \geq G \frac{\sqrt{d}}{\log d}$$

Proof. Note that $Y_d(s)$ and $X_d(s)$ have same sign in J , i.e $Y_d(s)$ has same sign as $\zeta(s)L(s, \chi)$.
since $L(s, \chi)$ has no zero in

$$1 - \frac{g}{\log d} \leq s \leq 1$$

$\zeta(s)L(s, \chi)$ doesn't change sign in that interval J . Since $X_d(s)$ is negative when s is sufficiently close to 1. $Y_d(s)$ is also negative when s is sufficiently close to 1, since we are given a zero free region, the sign of $Y_d(s)$ should not change in the interval and we get.

$$\begin{aligned} \Rightarrow Y_d(s) &\leq 0 \\ \Rightarrow \frac{c_1}{c_7}(1-s)d^{-(1-s)} - \frac{h'}{\sqrt{d}} &\leq 0 \quad \text{using } (*) \\ \Rightarrow \frac{h'}{\sqrt{d}} &\geq \frac{c_1}{c_7}(1-s)d^{-(1-s)} \end{aligned}$$

Define $\phi(s) = \frac{c_1}{c_7}(1-s)d^{-(1-s)}$ then ϕ attains a maximum of $\frac{c_1}{c_7} \frac{e^{-1}}{\log d}$ at $1 - \frac{1}{\log d}$ hence we get

$$\begin{aligned} \frac{h'}{\sqrt{d}} &\geq \max \phi(s) = \frac{c_1}{c_7} \frac{e^{-1}}{\log d} \\ \Rightarrow h' &\geq G \frac{\sqrt{d}}{\log d} \end{aligned}$$

□

Remark

Since $-d = b^2 - 4ac$ we get that $d \equiv 3, 0 \pmod{4}$. Note that when $d \equiv 0 \pmod{4}$, $(1, 0, d/4)$ gives us a reduced binary quadratic form with discriminant $-d$ and when $d \equiv 3 \pmod{4}$, $(1, 1, \frac{d+1}{4})$ works. i.e we always have a reduced binary quadratic form with discriminant $-d$ and $a = 1$. hence we get a lower bound for $\sum_Q \frac{1}{a} \geq 1$. Which gives us

$$h \geq h' \geq G \frac{\sqrt{d}}{\log d}$$

giving us a lower bound for class number. A lower bound for $\sum_Q \frac{1}{a}$ which depends on d is doesn't seem feasible, but if one is able to show that $\sum_Q \frac{1}{a} \gg \log d$ then we would have $h \gg d^{\frac{1}{2}}$ removing the epsilon from siegels Theorem.

References

- [1] K. Mahler, *On Hecke's theorem on the real zeros of the L -functions and the class number of quadratic fields*, Journal of the London Mathematical Society, Vol. 9 (1934), 299–305.
- [2] L. J. Mordell, *On the Riemann hypothesis and imaginary quadratic fields with a given class number*, Journal of the London Mathematical Society, Vol. 9 (1934), 289–298.
- [3] R. Meier and P. Seidel, *Binary Quadratic Forms and the Class Number Formula*, 2019.